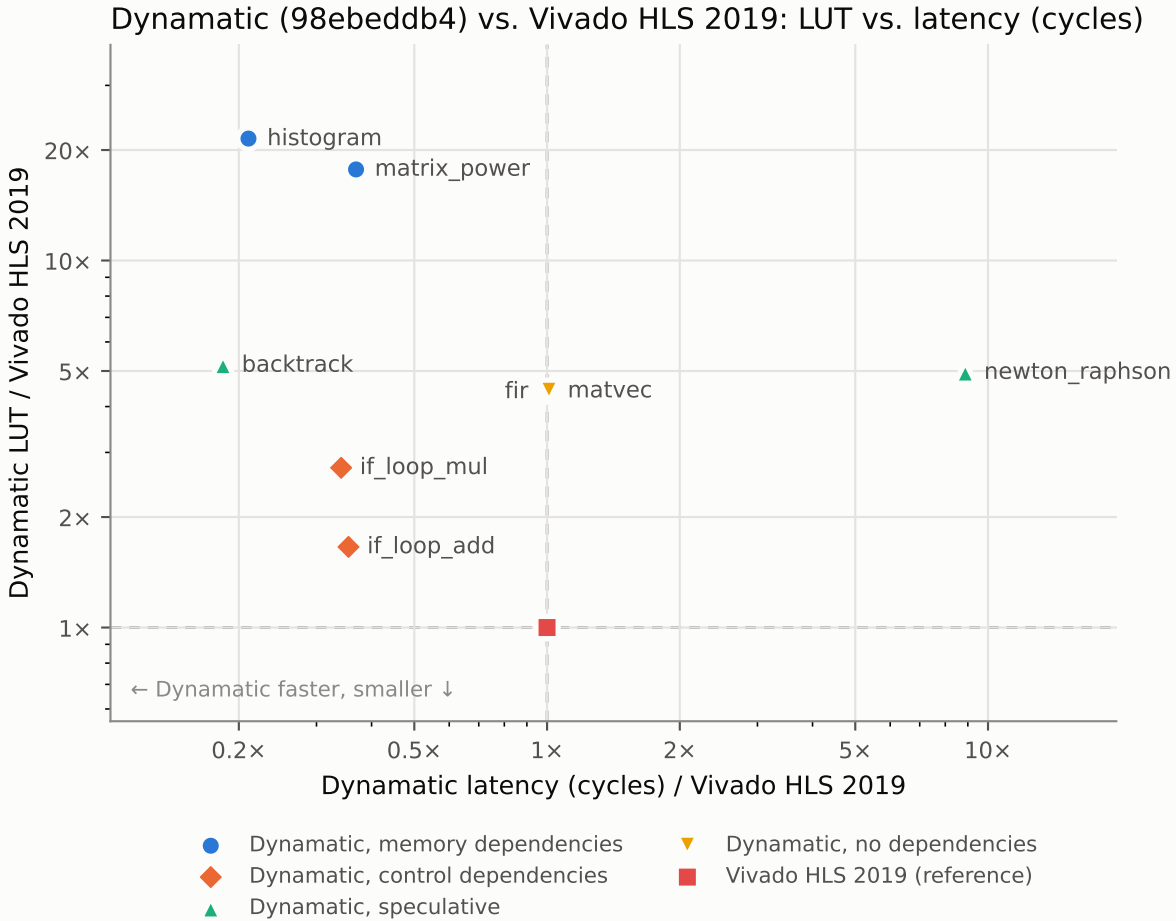
Dynamatic (98ebeddb4) vs. Vivado HLS 2019: LUT vs. latency (cycles)
Dynamatic LUT / Vivado HLS 2019
20×
10×
5×
2×
1×
histogram
matrix_power
backtrack
fir
matvec
newton_raphson
if_loop_mul
if_loop_add
← Dynamatic faster, smaller ↓
0.2×
0.5×
1×
2×
5×
10×
Dynamatic latency (cycles) / Vivado HLS 2019
Dynamatic, memory dependencies
Dynamatic, control dependencies
Dynamatic, speculative
Dynamatic, no dependencies
Vivado HLS 2019 (reference)